

POSITIVE-PARAMETER TWO-END SPATIAL DYNAMICS IN THE VAN DER POL REACTION–DIFFUSION SYSTEM

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ABSTRACT. We study stationary solutions of the one-dimensional van der Pol reaction–diffusion system in a singular positive-diffusion cusp. Assuming a frozen, publicly archived RFSN-II core and return–exit package, we prove on a nonempty compact positive parameter annulus that the stationary spatial system has an exhaustive high-winding first-return/first-exit relation. The two noncompact exits must be rebuilt from the full positive-parameter equations: one is a finite-spatial-distance pole with a Laurent–logarithmic action finite part, while the other is an infinite-distance algebraic end carried by a normally expanding future-staying hypersurface. We resolve the $K_2 \rightarrow K_1 \rightarrow$ outer matching corner, prove reference-subtracted length and action finite parts at the algebraic end, and attach both terminal potentials to the exact branch cocycle. The trapped return system is coded by a two-vertex countable shift and yields spatially periodic, localized multipulse, and bounded aperiodic stationary PDE profiles. The result concerns stationary spatial dynamics; it does not assert temporal stability, selection from a Turing bifurcation, or canard identification.

1. PROBLEM, GEOMETRY, AND CONDITIONAL MAIN RESULT

1.1. The physical stationary Hamiltonian. Consider

$$\begin{aligned} u_t &= v - f(u) + du_{xx}, \\ v_t &= \epsilon(a - u) + v_{xx}, \quad f(u) = \frac{1}{3}u^3 - u, \quad x \in \mathbb{R}, \end{aligned} \quad (1.1)$$

where $d > 0$ and $\epsilon > 0$. Its homogeneous state is $(u, v) = (a, f(a))$. Write $\delta = \sqrt{d}$ and introduce

$$\delta u_x = p, \quad \delta p_x = f(u) - v, \quad v_x = q, \quad q_x = \epsilon(u - a). \quad (1.2)$$

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Stationary profiles of (1.1) are precisely the complete spatial orbits of (1.2). With

$$\begin{aligned} F(u) &= \frac{1}{12}u^4 - \frac{1}{2}u^2, \\ \mathcal{G}_\delta &= \frac{1}{2}(\epsilon p^2 - q^2) - \epsilon\{F(u) + (a - u)v\}, \\ \lambda_\delta &= \epsilon p du - \delta^{-1}q dv, \end{aligned} \quad (1.3)$$

direct differentiation gives $d\mathcal{G}_\delta/dx = 0$. Moreover, if $\omega_\delta = d\lambda_\delta$, then

$$\iota_{X_x}\omega_\delta = dH_{\text{ph}}, \quad H_{\text{ph}} = -\frac{\mathcal{G}_\delta - \mathcal{G}_\delta(O)}{\delta}, \quad O = (a, 0, f(a), 0). \quad (1.4)$$

The reverser

$$\mathcal{R}(u, p, v, q) = (u, -p, v, -q) \quad (1.5)$$

is anti-symplectic. Thus the zero-energy stationary spatial problem is a reversible exact Hamiltonian system. The sign and clock in (1.4) matter for every action formula below.

The model and the singular Turing-canard regime were analyzed in [6]. We use its physical equations and blow-up charts, but not a positive-parameter noncompact-end theorem: neither end used here is obtained by ordinary persistence from the singular core. The saddle-focus homoclinic mechanism behind the return geometry is classical [1, 4]; the issue addressed here is the full positive-parameter attachment of that compact mechanism to two inequivalent singular exits of (1.2).

1.2. The cusp and the two terminal geometries. Fix

$$0 < \epsilon_- < \epsilon_+ < \infty, \quad A > 0, \quad (1.6)$$

and parameterize the positive cusp by

$$d = r^4, \quad \delta = r^2, \quad a = 1 + \sqrt{\epsilon} r^3 a_2, \quad |a_2| \leq A. \quad (1.7)$$

For small r , the homogeneous state is a saddle-focus for the spatial system. A source point that spends a long spatial time near this equilibrium makes many rotations before meeting the global event block. It may return to the source, meet a finite auxiliary boundary, enter a pole at which $u \rightarrow +\infty$ at a finite physical position, or enter an algebraic channel that reaches $u = +\infty$ only as $x \rightarrow +\infty$.

These last two outcomes are genuinely different. At the pole, the physical remaining distance $s = x_b - x$ is the natural boundary variable and the action requires a Laurent–logarithmic subtraction. At the algebraic end, $z = u^{-1}$ is the boundary defining function, physical length diverges, and both length and action are normalized against a complete reference tail. The geometric relation is summarized in Figure 1.

SCHEMATIC: stationary spatial dynamics; geometry and distances are not quantitative
finite auxiliary exits

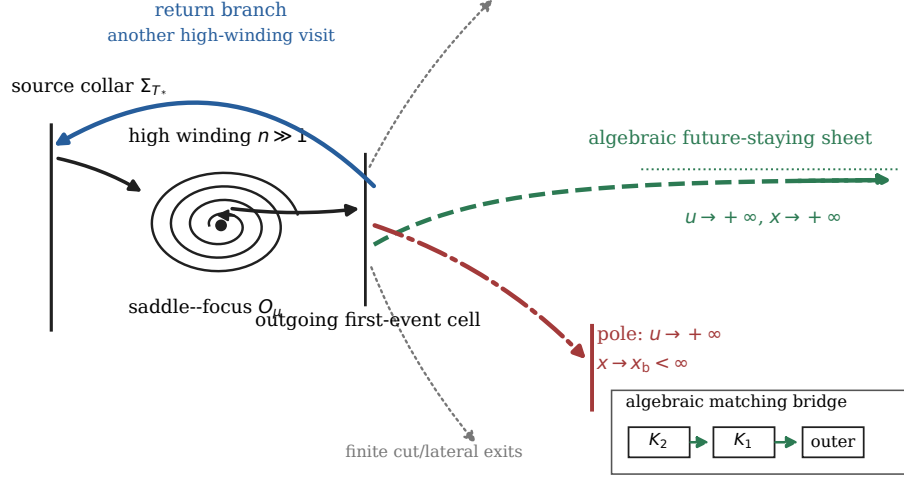


FIGURE 1. Schematic stationary spatial geometry. A point in the physical source collar makes a high-winding saddle-focus passage and then returns, reaches the finite-distance pole, reaches the infinite-distance algebraic end, or meets a named finite boundary. The inset identifies the $K_2 \rightarrow K_1 \rightarrow \text{outer}$ bridge needed to attach the algebraic event. Curve placement and distances are qualitative. The figure does not represent temporal stability, Turing selection, or canard tracking.

1.3. Frozen input. The application theorem is conditional on a precisely delimited source package. This is preferable to hiding a computer-assisted core and a fixed-system return theorem behind the phrase “standard theory.” The immutable source, certificates, environment, and replay manifest are included in the public release cited in [5]; hashes and replay commands are placed in the accompanying provenance supplement.

Throughout the paper, a mixed C^k estimate is taken on the displayed fixed state domains and means a uniform bound for all state/parameter derivatives $D_Z^i D_\mu^j$ with $i + j \leq k$ and $j \leq 2$. Weighted saddle estimates may also contain any fixed number of logarithmic derivatives $D_{\log \nu} = \nu \partial_\nu$.

Hypothesis 1.1 (Frozen RFSN-II package). The following data and analytic implication hold at the singular core and in the associated fixed-system construction.

(H_{core}) The reversible exact Hamiltonian core

$$U' = P, \quad P' = -U^2 - V, \quad V' = Q, \quad Q' = U \quad (1.8)$$

has the selected symmetric homoclinic orbit, transverse modulo the flow in the regular zero-energy hypersurface. Its true unstable source graph and one compact physical event block have certified first-hit

assignments, clean and neat incidences, exhaustive connected sign strata, fixed corner priorities, and strictly positive rank, speed, separation, and ordering margins. The block contains distinct homoclinic, algebraic-directed, and pole-directed anchors. Local uniqueness of the homoclinic is required only in the frozen shooting tube. More specifically, on the nonempty pole phase interval $I_p = [-0.2, 0.2]$, put

$$x = -U, \quad y = -P, \quad z_c = -V, \quad \zeta = -Q, \quad D = \frac{1}{2}x^2 - z_c, \quad K = xy - \zeta.$$

Every frozen source orbit in I_p has a strict first hit of $x = 10$ in $\{y > 0, D > 0, K > 0\}$, with uniform positive hit-speed, cone-entry, and earlier-event margins.

(H_{Jost}) On a fixed central section $h_c = 0$, the singular comparison sheet has a defining function g_a . If $y_0(\phi, t) = \Phi_0^t(S_0(\phi))$ is the source flight to its selected intersection, then

$$s_0 = dh_c(F_0)(y_0) \neq 0, \quad \chi_0 = Dg_a(y_0)D_\phi y_0 \neq 0. \quad (1.9)$$

The same sheet has an endpoint-normalized adjoint/Jost row ψ and a fixed growing complement $\mathbf{u}_0$ whose directed exchange pairing is

$$\chi_{\text{ex}}^0 = \psi(\mathbf{u}_0) = 144\sqrt{3}. \quad (1.10)$$

(H_{ret}) The frozen mixed-passage, selector, first-exit-family, covariance, and terminal-extension results are available with the following input–output contract. Their inputs are: a compact C^2 family of reversible exact saddle-foci with an exact transverse-action passage; a transverse compact homoclinic transition with a fixed limiting matching rectangle, inverse and range margins; a finite state- C^3 , parameter- C^2 clean physical event arrangement with strictly positive rank, speed, buffer, and ordering margins; and terminal potentials with one spare state derivative. For

$$T = -\alpha_\mu^{-1} \log |\nu| + t_{\mu,\sigma} + R_T, \quad \Delta = -\frac{\beta_\mu}{\alpha_\mu} \log |\nu| + b_{\mu,\sigma} + R_\Delta,$$

the remainders and their mixed two-jets are $O(|\nu|(1 + |\log |\nu||))$ in $D_{\log \nu}$. The event input includes every competing physical event and every pairwise event-time difference: an empty tie class has a positive gap, while a nonempty tie has nonzero conormal and a fixed priority. Under these inputs the frozen results produce one uniform winding selector, completed contracting cross forms, the literal clean first-event census, exact section/gauge covariance, and the two-vertex countable terminal coding theorem with mixed two-jet estimates.

Hypothesis 1.1 imports neither positive-parameter end. The pole, outer future-staying hypersurface, their matching, and both terminal finite parts are all constructed below from (1.2). The relative invariant-manifold statement used at the resolved K_1 corner and the finite marked-atlas descent are proved in the companion proof archive, using the classical compact invariant-manifold theory of [2, 3] as boundaryless input.

1.4. Main theorem.

Theorem 1.2 (Conditional positive-parameter two-end theorem). *Assume Hypothesis 1.1. For the constants in (1.6), there exists $r_V > 0$ such that all of the following statements hold uniformly on the nonempty compact positive annulus*

$$\mathcal{P}_V = \left[\frac{1}{2}r_V, r_V \right] \times [-A, A] \times [\epsilon_-, \epsilon_+], \quad \mu = (r, a_2, \epsilon), \quad (1.11)$$

with physical parameters given by (1.7).

- (1) *The homogeneous state is a uniform saddle-focus, and the selected symmetric zero-energy homoclinic, its local stable and unstable manifolds, its exact local passage, and the compact central first-event block continue with two derivatives in μ in the blown-up parameter coordinates.*
- (2) *There is a physical source collar Σ_{T^*} and a finite compatible marked cover on which every sufficiently high-winding point has exactly one compact first event: return, stable cut, a named finite lateral event, or a terminal carrier labelled by the pole or algebraic outcome. Every point on either labelled terminal carrier has the stated subsequent noncompact future. The connected strata form a literal disjoint census with no unnamed residual component. For every sufficiently large local winding, both return target signs occur and the anchored pole and algebraic components are nonempty. No nonemptiness is asserted for every other source sign or component label. On chart overlaps, the physical first event agrees, while raw winding integers may differ by a bounded recoding.*
- (3) *The pole occurs at a finite physical position x_b and has a mixed- C^2 renormalized action*

$$\mathcal{A}_p^{\text{ren}} = \lim_{s \downarrow 0} \left[\int_{x_0}^{x_b - s} \lambda_\delta - 2\epsilon\delta^3 s^{-3} + 2\epsilon\delta s^{-1} - \sqrt{6}\epsilon Z_0 \log s \right]. \quad (1.12)$$

where x_0 is the fixed source cut in the chosen marking; moving it changes the terminal potential by exactly the intervening finite branch action. The algebraic branch reaches $u = +\infty$ only at infinite physical distance. In the coordinate $Q = z^{-2}$ it has mixed- C^2 renormalized length and action

$$\begin{aligned} L_a^{\text{ren}} &= \lim_{Q \rightarrow \infty} \{L_{\text{source} \rightarrow Q} - C_{x,\mu}(Q)\}, \\ \mathcal{A}_a^{\text{ren}} &= \lim_{Q \rightarrow \infty} \left\{ \int_{\text{source} \rightarrow Q} \lambda_\delta - C_{A,\mu}(Q) \right\}, \end{aligned} \quad (1.13)$$

where the counterterms are complete reference-tail integrals and satisfy

$$\begin{aligned} C_{x,\mu}(Q) &= q_*^{-1} Q^{1/2} + O_{C_\mu^2}(\log Q), \\ C_{A,\mu}(Q) &= -\frac{q_*}{5\delta} Q^{5/2} + O_{C_\mu^2}(Q^2), \quad q_* = \sqrt{\epsilon/2}. \end{aligned} \quad (1.14)$$

All finite branch actions and the two terminal potentials obey exact first-hit composition; closed actions are independent of admissible cuts, exact gauges, event-free section slides, and the local marking.

- (4) *In every marking, the trapped return set is conjugate to the two-vertex countable edge shift with edges*

$$(\sigma, n, \sigma'), \quad \sigma, \sigma' \in \{+, -\}, \quad n \geq N. \quad (1.15)$$

Primitive periodic words yield smooth spatially periodic stationary solutions of (1.1); nonperiodic bi-infinite words yield bounded non-periodic stationary solutions; and arbitrary finite admissible edge words with the selected unstable and stable endpoints yield homoclinic multipulse profiles converging to $(a, f(a))$ as $x \rightarrow \pm\infty$.

The set (1.11) is existential. It is not the currently proposed numerical box, and the theorem does not claim uniformity down to the cusp tip $r = 0$. The parameter image in (d, a, ϵ) is a cusp, not a rectangular box.

For completeness, the quantitative high-winding consequence is stated next. Let

$$c_\mu = 2ra_2 + \sqrt{\epsilon} r^4 a_2^2, \quad \alpha_\mu = \frac{1}{2} \sqrt{2 + c_\mu}, \quad \beta_\mu = \frac{1}{2} \sqrt{2 - c_\mu}. \quad (1.16)$$

For any

$$0 < \varkappa < \inf_{\mu \in \mathcal{P}_V} \frac{2\pi\alpha_\mu}{\beta_\mu}, \quad (1.17)$$

the marked cover and N may be chosen so that its cross forms and branch potentials converge in mixed C^2 at rate $C(1+n)^3 e^{-\varkappa n}$. In particular, for a fixed primitive cyclic word with windings $n_0, \dots, n_{m-1}$,

$$L(\gamma_{\mathbf{n}}) = \frac{2\pi r}{\epsilon^{1/4} \beta_\mu} \sum_{j=0}^{m-1} n_j + \mathcal{L}_{\sigma, \mu} + O\left(\max_j (1+n_j)^3 e^{-\varkappa n_j}\right), \quad (1.18)$$

while its closed action has a finite marking-independent limit with the same error. The estimate is for fixed word length; it is not uniform when the number of composed branches tends to infinity. Here σ denotes the cyclic sequence of source and target signs of the fixed word, so the finite offset need not be a function of a single sign.

1.5. Why the proof is not ordinary persistence. The frozen theory reaches the compact central gates. It does not identify their positive-parameter futures. The proof therefore has the following causal spine:

$$\begin{aligned}
& \text{physical Hamiltonian and frozen core} \longrightarrow \text{central continuation,} \\
& \text{pole-directed gate} \longrightarrow \text{positive pole and its finite part,} \\
& \text{positive outer field} \longrightarrow \text{future-staying sheet,} \\
& \text{central gate} + \text{outer sheet} + (\mathbf{H}_{\text{Jost}}) \longrightarrow K_2 \rightarrow K_1 \rightarrow \text{outer matching,} \\
& \text{matched sheet} \longrightarrow \text{algebraic length/action finite parts,} \\
& \text{compact event block} + \text{two actual terminals} + (\mathbf{H}_{\text{ret}}) \\
& \longrightarrow \text{physical first-event relation} \longrightarrow \text{stationary PDE profiles.}
\end{aligned} \tag{1.19}$$

The pole changes dominant balance when $\delta > 0$. The algebraic branch enters an outer slow regime and must be matched across a nonhyperbolic chart corner. These are the two obstructions removed in Sections 2 and 3; Section 4 then performs only the minimal replacement of the two finite gates by the actual terminals.

2. CENTRAL CONTINUATION AND THE FINITE-DISTANCE POLE

2.1. Exact central scaling. The cusp scaling is fixed before applying the imported core. Put

$$\begin{aligned}
u &= a - \sqrt{\epsilon} r^2 U, & p &= -\epsilon^{3/4} r^3 P, \\
v &= f(a) - \epsilon r^4 V, & q &= -\epsilon^{5/4} r^3 Q, & \xi &= \epsilon^{1/4} x/r.
\end{aligned} \tag{2.1}$$

This sends the homogeneous state to the origin. Direct substitution, with no remainder, gives

$$\begin{aligned}
U' &= P, \\
P' &= c_\mu U - V - (1 + \sqrt{\epsilon} r^3 a_2) U^2 + \frac{\sqrt{\epsilon}}{3} r^2 U^3, \\
V' &= Q, & Q' &= U,
\end{aligned} \tag{2.2}$$

where the prime denotes $d/d\xi$ and c_μ is defined in (1.16). The parameter-independent primitive and shifted Hamiltonian are

$$\begin{aligned}
\lambda &= P dU - Q dV, \\
\hat{H}_\mu &= \frac{1}{2}(Q^2 - P^2) - UV + \frac{c_\mu}{2} U^2 - \frac{1 + \sqrt{\epsilon} r^3 a_2}{3} U^3 + \frac{\sqrt{\epsilon}}{12} r^2 U^4, \\
\iota_{F_\mu} d\lambda &= d\hat{H}_\mu.
\end{aligned} \tag{2.3}$$

The exact physical conversions are

$$dx = r\epsilon^{-1/4} d\xi, \quad \lambda_\delta = \epsilon^{9/4} r^5 \lambda, \quad H_{\text{ph}} = \epsilon^{5/2} r^4 \hat{H}_\mu. \tag{2.4}$$

At $r = 0$, (2.2) is exactly (1.8). Its characteristic polynomial at the origin is $\zeta^4 - c_\mu \zeta^2 + 1$, and hence its spectrum is $\{\pm\alpha_\mu \pm i\beta_\mu\}$.

Proposition 2.1 (Central continuation). *Under (H_{core}) and the mixed-passage clause of (H_{ret}) , after decreasing the upper r -radius, the origin, the selected symmetric homoclinic, and the compact central first-event block continue on the closed blown-up parameter wedge. The local invariant manifolds and the homoclinic have two parameter derivatives with uniform exponential tails. A finite parameter cover carries reversible exact saddle coordinates. If ν is the signed transverse action, the passage time and Kato-oriented phase satisfy*

$$\begin{aligned} T_{\mu,\sigma}(\nu) &= -\alpha_\mu^{-1} \log |\nu| + t_{\mu,\sigma} + O_{\text{wl}}(|\nu|), \\ \Delta_{\mu,\sigma}(\nu) &= -\frac{\beta_\mu}{\alpha_\mu} \log |\nu| + b_{\mu,\sigma} + O_{\text{wl}}(|\nu|), \end{aligned} \quad (2.5)$$

where $O_{\text{wl}}(|\nu|)$ denotes $O(|\nu|(1 + |\log |\nu||))$ after any two parameter derivatives and any fixed number of logarithmic derivatives $\nu \partial_\nu$. The continued event block remains clean, neat, componentwise exhaustive, and separated by uniform positive physical first-hit margins.

Proof. On fixed compact state sets, (2.2) is a smooth C^2 parameter family converging to (1.8). The saddle-focus rates stay uniformly away from zero. Parameter-dependent graph transforms give fixed domain stable and unstable graphs; their Lyapunov–Perron equations may be differentiated twice because the exponential dichotomy is uniform. In a weighted full-line space, the reversible matching equation for the selected core orbit has invertible derivative by the certified transversality in (H_{core}) . The implicit-function theorem therefore gives the homoclinic and its two-derivative exponential tails.

The exact local passage follows from the reversible analytic saddle normal form in (H_{ret}) after tangent normalization to one Kato-transported oriented spectral frame. Direct normal-form quadrature on the radial sections fixes the minus sign in the phase law in (2.5); the transverse action itself is unchanged. Finally, all compact first-hit speeds, active conormal ranks, empty incidences, earlier-event exclusions, and ordering inequalities in (H_{core}) are strict on compact sets. Smooth first-hit dependence and a boundary-preserving controlled isotopy retain the complete labelled incidence complex. This proves the proposition. $\square$

The two end labels in Proposition 2.1 are still finite gates. Nothing in compact first-hit stability implies that an orbit subsequently reaches a noncompact end.

2.2. Invariant cone and finite physical blow-up. For the pole-directed branch write

$$x = -U, \quad y = -P, \quad z = -V, \quad \zeta = -Q, \quad (2.6)$$

and set

$$D = \frac{1}{2}x^2 - z, \quad K = xy - \zeta, \quad \mathcal{K}_\mu = \{x \geq 10, y > 0, D \geq 0, K \geq 0\}. \quad (2.7)$$

The exact field gives

$$\begin{aligned} x' &= y, & D' &= K, \\ y' &= D + (B - \tfrac{1}{2})x^2 + c_\mu x + bx^3, & K' &= y^2 + xy' - x. \end{aligned} \quad (2.8)$$

where $B = 1 + \sqrt{\epsilon}r^3a_2$ and $b = \sqrt{\epsilon}r^2/3 > 0$. On the final small positive annulus, $B \geq 3/4$ and $|c_\mu| \leq 1$. Hence the faces of $\mathcal{K}_\mu$ point inward and, for $x \geq 10$,

$$y' > 0, \quad K' > 0, \quad y \frac{dy}{dx} \geq bx^3. \quad (2.9)$$

The certified core source window has a strict first hit of $x = 10$ in the interior of this cone. Proposition 2.1 and a short event-free flowbox slide retain that literal first hit for positive parameters. From (2.9),

$$y(x)^2 \geq y(10)^2 + \frac{b}{2}(x^4 - 10^4), \quad \xi_b - \xi = \int_x^\infty \frac{dX}{y(X)} < \infty. \quad (2.10)$$

Thus the pole is reached in finite central and physical spatial time. On the zero-energy level the exact identity is

$$y^2 = \zeta^2 - 2xz + c_\mu x^2 + \frac{2B}{3}x^3 + \frac{b}{2}x^4. \quad (2.11)$$

The lower bound in (2.10) gives

$$\frac{d\zeta}{dx} = \frac{x}{y} = O(x^{-1}), \quad \frac{dz}{dx} = \frac{\zeta}{y} = O((1 + \log x)x^{-2}).$$

Hence $z \rightarrow z_\infty$, $\zeta = O(1 + \log x)$, and (2.11) closes the uniform limits

$$\frac{y}{x^2} \rightarrow \sqrt{b/2}, \quad \zeta - \sqrt{2/b} \log x \rightarrow \zeta_\infty, \quad z_\infty - z = O((1 + \log x)x^{-1}). \quad (2.12)$$

One further integration of $dx/d\xi = y$ and the exact physical inverse of (2.1) give the balance below. In particular, the two end labels range in a compact set on the transported source window.

2.3. Regular-singular compactification and pole action. Let $s = x_b - x$ and $\ell = \sqrt{6}\delta$. The sharp balance obtained from the conserved energy and (2.9) is

$$\frac{su}{\ell} \rightarrow 1, \quad \frac{s^2p}{\ell\delta} \rightarrow 1, \quad q + \ell\epsilon \log s \rightarrow W_0, \quad v - \ell\epsilon s \log s \rightarrow Z_0. \quad (2.13)$$

Define

$$X = su/\ell, \quad Y = s^2p/(\ell\delta), \quad W = q + \ell\epsilon \log s, \quad Z = v - \ell\epsilon s \log s, \quad (2.14)$$

and use $\dot{} = s d/dx$. Substitution in (1.2) gives the exact C^3 regular-singular field

$$\begin{aligned} \dot{s} &= -s, & \dot{X} &= -X + Y, \\ \dot{Y} &= -2Y + 2X^3 - s^2\delta^{-2}X - s^3(\ell\delta^2)^{-1}Z - \epsilon\delta^{-2}s^4\log s, \\ \dot{W} &= \ell\epsilon(X - 1) - a\epsilon s, & \dot{Z} &= s(W + \ell\epsilon). \end{aligned} \tag{2.15}$$

At $s = 0$ it has the two-dimensional equilibrium manifold $X = Y = 1$, parameterized by (Z_0, W_0) . Its normal spectrum is $\{-1, -4, +1\}$, while the two label directions have eigenvalue zero. The stable pole family therefore has normalized power spectrum

$$\{-1, 0, 0, 1, 4\}. \tag{2.16}$$

Proposition 2.2 (Positive pole terminal). *Under Hypothesis 1.1, on a nonempty compact positive subannulus, the transported pole source window enters an open pole basin and reaches $u = +\infty$ at a finite physical position. Each pole orbit has unique labels (Z_0, W_0, c_4) and a polyhomogeneous expansion with admissible powers 1 and 4, including the forced $s\log s$ and $s^4(\log s)^2$ terms. On the fixed zero-energy level, c_4 is determined uniquely because*

$$\partial_{c_4}\mathcal{G}_\delta = -30\epsilon\delta^4 \neq 0. \tag{2.17}$$

The action finite part (1.12) exists with two mixed parameter derivatives and satisfies exact moving-cut additivity.

Proof. Normal hyperbolicity of the compact label rectangles in (2.15) gives a four-dimensional local stable set: two label directions and two stable fibers. The uniform limits (2.13) and (2.12) show that every transported source orbit converges to the interior of one common compact label rectangle. The defining staying-set characterization of the local stable manifold therefore puts the entire window in that stable set. Eliminating Y yields the scalar Fuchsian equation

$$(s\partial_s - 4)(s\partial_s + 1)(X - 1) = 6(X - 1)^2 + 2(X - 1)^3 - \frac{s^2}{\delta^2}X - \frac{s^3}{\ell\delta^2}Z - \frac{\epsilon}{\delta^2}s^4\log s. \tag{2.18}$$

The nonresonant powers two and three are forced; at the indicial root four, the constant and logarithmic forcing produces $s^4\log s$ and $s^4(\log s)^2$, with c_4s^4 the free stable mode. The Green inverse of $(s\partial_s - 4)(s\partial_s + 1)$ is bounded on the conormal weight $s^5(1 + |\log s|)^2$. A uniform contraction and two differentiated fixed-point equations prove the expansion and the mixed regularity. The physical inverse of (2.14) has nonzero leading Jacobian, so these are genuine local state coordinates and their image is an open pole basin. Substitution into (1.3) gives (2.17).

Finally,

$$\lambda_\delta(\partial_x) = 6\epsilon\delta^3s^{-4} - 2\epsilon\delta s^{-2} - \sqrt{6}\epsilon Z_0s^{-1} + O_{C^2}((1 + |\log s|)^2). \tag{2.19}$$

The remainder is integrable uniformly after two mixed derivatives. Integrating (2.19) proves (1.12). Moving a finite cut merely transfers the same finite orbit integral between the central and terminal terms, proving exact additivity. $\square$

3. THE ALGEBRAIC END: OUTER GEOMETRY, MATCHING, AND FINITE PARTS

3.1. A positive-parameter future-staying hypersurface. The algebraic channel is most transparent in the fast clock $y = x/\delta$. For $u > 0$ set

$$z = u^{-1}, \quad \pi = p, \quad w = z\{f(u) - v\}, \quad \chi = z^2 q, \quad \frac{d}{d\tau} = z \frac{d}{dy}. \quad (3.1)$$

The exact compactified field is

$$\begin{aligned} \dot{z} &= -\pi z^3, & \dot{\pi} &= w, \\ w &= (1 - z^2)\pi - \delta\chi - \pi w z^2, & \dot{\chi} &= z^2\{\epsilon\delta(1 - az) - 2\pi\chi\}. \end{aligned} \quad (3.2)$$

Diagonalize the fast normal pair by

$$h = \pi - \delta\chi, \quad \alpha = (h + w)/2, \quad \beta = (h - w)/2. \quad (3.3)$$

Thus $\pi = \delta\chi + \alpha + \beta$ and $w = \alpha - \beta$. The shifted energy

$$E = -\mathcal{G}_\delta + \mathcal{G}_\delta(O) = -\mathcal{G}_\delta - \epsilon F(a) \quad (3.4)$$

is a regular boundary coordinate. In the outer variables its exact energy equation is

$$\begin{aligned} \chi^2 &= \frac{\epsilon}{2} - \frac{2a\epsilon}{3}z - \epsilon(2w + 1)z^2 + 2a\epsilon(w + 1)z^3 \\ &+ \{\epsilon\pi^2 + 2E + 2\epsilon F(a)\}z^4. \end{aligned} \quad (3.5)$$

At $z = 0$ the derivative with respect to the positive χ root is $2\sqrt{\epsilon/2} > 0$. Hence the implicit-function theorem gives

$$\chi = \text{Chi}(z, E, \beta, \alpha; \mu) > 0, \quad \text{Chi}(0, E, \beta, \alpha; \mu) = q_*(\epsilon) = \sqrt{\epsilon/2}, \quad (3.6)$$

and the boundary field in (z, E, β, α) is exactly

$$(\dot{z}, \dot{E}, \dot{\beta}, \dot{\alpha}) = (0, 0, -\beta, \alpha). \quad (3.7)$$

Thus β is future stable and α future unstable. The positive cubic balance has already been absorbed in the regular root (3.6); there is no coefficient singular in r at $z = 0$.

Proposition 3.1 (Outer future-staying graph). *On the final compact positive annulus, there is a fixed corridor in (z, E, β, α) whose maximal forward-staying set is a unique codimension-one graph*

$$\mathcal{W}_{\text{out}, \mu}^{\text{tail}} = \{\alpha = \Gamma_\mu(z, E, \beta)\}. \quad (3.8)$$

It is a locally maximal C^3 manifold with corners, normally expanding in forward τ -time, third-order bunched, and has mixed regularity

$$\Gamma \in C_\mu^0 C_{z, E, \beta}^3 \cap C_\mu^1 C_{z, E, \beta}^2 \cap C_\mu^2 C_{z, E, \beta}^1. \quad (3.9)$$

Every orbit on the zero-energy future sheet with $z > 0$ satisfies

$$\begin{aligned} u_x &= q_* + O(u^{-1}), & q &= q_* u^2 + O(u), \\ v &= f(u) + O(u^{-1}), & p &= \delta q_* + O(u^{-1}). \end{aligned} \quad (3.10)$$

and reaches $z = 0$ only as $x \rightarrow +\infty$.

Proof. Choose a product corridor so that $z = 0$ and the energy faces are invariant, the remaining base faces point inward, and the two α faces are strict exits. By (3.7), after decreasing the fixed outer radius, the base logarithmic norm and the two off-diagonal derivative blocks are at most ν , while the normal expansion is at least $1 - \nu$. The tangent and normal quotient cocycles for one fixed time $T > 0$ then obey

$$\sup_{\mu, Z} \left\| (N_{\mu, Z}^T)^{-1} \right\| \left\| A_{\mu, Z}^T \right\|^j < 1, \quad j = 0, 1, 2, 3. \quad (3.11)$$

A graph transform on a doubled collar, restricted back using the inward and invariant face signs, gives the unique staying graph and (3.9). This is the relative-with-corners form of the standard compact invariant-manifold argument [2, 3]; the collar restriction and its fixed-extension uniqueness are part of the companion proof archive.

On the graph, $d(z^{-2})/d\tau = 2\pi$ with π uniformly positive. Bounded variation of constants removes the growing α mode and gives $\alpha, \beta = O(z^2)$. The energy root then yields $\chi = q_* + O(z)$ and $\pi = \delta q_* + O(z)$. Substitution in (3.1) proves (3.10); since $dx/d\tau = \delta z$ and $z(\tau) \asymp (1 + \tau)^{-1/2}$, the physical length diverges. $\square$

3.2. Resolution of the $K_2 \rightarrow K_1 \rightarrow$ outer corner. Proposition 3.1 does not imply that the finite algebraic-directed gate from Proposition 2.1 lies on (3.8). The gate lives in the universal central chart K_2 ; the future sheet is constructed in the physical outer chart; and the two meet through the entry/exit chart K_1 . The overlap degenerates as $r \rightarrow 0$, so a compact-away-from-the-corner Fenichel argument does not control the two parameter derivatives needed by the return theorem.

The exact K_2 – K_1 transition on the positive overlap is

$$\begin{aligned} r_1 &= r\sqrt{u_2}, & \delta_1 &= u_2^{-1}, \\ p_1 &= u_2^{-3/2} p_2, & v_1 &= u_2^{-2} v_2, \\ q_1 &= u_2^{-3/2} q_2, & a_1 &= u_2^{-3/2} a_2, \end{aligned} \quad (3.12)$$

with $dy_2 = u_2^{-1/2} dy_1$. We resolve the corner on a closed comparison family first, choose all smallness thresholds there, and only afterward freeze a positive annulus.

The outer trace remains regular at the comparison face after setting

$$A = \alpha/\delta, \quad B = \beta/\delta, \quad H = \frac{E}{\epsilon^{5/2} \gamma^6}, \quad S = \chi + A + B, \quad D_o = A - B. \quad (3.13)$$

The exact scaled normal field is

$$\begin{aligned}\dot{z} &= -\delta S z^3, & \dot{H} &= 0, \\ \dot{A} &= A - \frac{1}{2} z^2 S + \frac{1}{2} \delta z^2 \{-\epsilon(1 - az) + 2\chi S - S D_o\}, \\ \dot{B} &= -B + \frac{1}{2} z^2 S + \frac{1}{2} \delta z^2 \{-\epsilon(1 - az) + 2\chi S + S D_o\}.\end{aligned}\tag{3.14}$$

Its positive energy root is smooth through $\delta = 0$. With $C = A + B$, the normal subsystem there is

$$\dot{C} = D_o, \quad \dot{D}_o = (1 - z^2)C - z^2 \chi_0(z),$$

and has rates $\pm\sqrt{1 - z^2}$. On $0 \leq z \leq z_* < 1$, adapted quotient metrics give, for some $\theta > 0$,

$$\mu_2(C_{\tan}) \leq \theta, \quad \|B_{\text{cr}}\|, \|D_{\text{cr}}\| \leq \theta, \quad a_n \geq \sqrt{1 - z_*^2} - \theta, \quad 8\theta < \sqrt{1 - z_*^2}.\tag{3.15}$$

The corner itself is regular in

$$s_\epsilon = \sqrt{\epsilon}, \quad \sigma = \sqrt{\delta_1}, \quad \Pi = p_1/\delta_1, \quad \Omega = \frac{1 - v_1 + (s_\epsilon/3)r_1^2}{\delta_1}, \quad r = r_1\sigma.\tag{3.16}$$

On the positive energy root, the exact K_1 field is

$$\begin{aligned}r_1' &= \frac{s_\epsilon}{2} \sigma^2 \Pi r_1, & \sigma' &= -\frac{s_\epsilon}{2} \sigma^3 \Pi, \quad H' = 0, \\ \Pi' &= \Omega - \frac{s_\epsilon}{2} \sigma^2 \Pi^2, \\ \Omega' &= (2s_\epsilon + \epsilon r_1^2) \Pi - s_\epsilon q_1 - s_\epsilon \sigma^2 \Pi \Omega, \\ q_1' &= \sigma^2 \{1 - r_1 \sigma^3 a_2 - \frac{3}{2} s_\epsilon \Pi q_1\}.\end{aligned}\tag{3.17}$$

Here q_1 is the unique positive root of the regular energy equation; its implicit derivative is $12s_\epsilon q_1 > 0$. At $\sigma = 0$,

$$q_{10} = \sqrt{\frac{8 + 3s_\epsilon r_1^2}{6s_\epsilon}}, \quad \Pi_0 = \frac{q_{10}}{2 + s_\epsilon r_1^2}, \quad \Omega_0 = 0,$$

and the normal rates are $\pm\lambda_1$, where

$$\lambda_1 = \sqrt{s_\epsilon(2 + s_\epsilon r_1^2)} \geq \lambda_* := \sqrt{2} \epsilon_-^{1/4}.$$

On the fixed cylinder $0 \leq r_1 \leq R$, $0 \leq \sigma \leq \sigma_0$, the collars can therefore be chosen so that the order- j quotient gaps obey

$$\gamma_j \geq \lambda_* - (2j + 2)\theta > 0, \quad 0 \leq j \leq 3.\tag{3.18}$$

The exact transition to (3.13) is

$$\begin{aligned}z &= (1 + s_\epsilon r_1^2)^{-1}, & \chi &= z^2 \epsilon^{3/2} r_1^3 q_1, \\ A + B &= \epsilon r_1 \Pi - \chi, & A - B &= \epsilon z r_1^2 \Omega.\end{aligned}\tag{3.19}$$

The invariant identity $(r_1\sigma)' = 0$ and $\Pi_0 > 0$ give a directed base crossing from the K_2 overlap to the outer overlap. The relative overflowing invariant-manifold theorem applied on doubled collars, followed by the common-cone gluing allowed by (3.15)–(3.18), constructs the codimension-one graph through the entire block with three state jets and two parameter derivatives.

Proposition 3.2 (Central–outer matching). *Under Hypothesis 1.1, after one final choice of the positive annulus, a fixed patch of (3.8) has a unique backward saturation through the outer tube, resolved K_1 corner, and K_2 overlap. Its trace on a fixed central section is a cooriented C^3 graph*

$$\mathcal{W}_{\text{out},\mu}^{\text{match}}. \quad (3.20)$$

If G_0 is the frozen singular comparison graph and G_μ the resolved trace, then

$$\|G_\mu - G_0\|_{C^2} \leq Cr, \quad \sup_{i+j \leq 2} \|D_Z^i D_\mu^j G_\mu\| < \infty. \quad (3.21)$$

Under (H_{Jost}) , the endpoint exchange pairing satisfies

$$\chi_{\text{ex},\mu} = 144\sqrt{3} + O(r) \geq 72\sqrt{3}. \quad (3.22)$$

This is distinct from the source matching determinant. If $y_\mu(\phi, t) = \Phi_\mu^t(S_\mu(\phi))$ and $\bar{g}_{c,\mu}$ defines the matched target on $h_c = 0$, put

$$\mathfrak{M}_\mu(\phi, t) = \begin{pmatrix} \tilde{g}_{c,\mu}(y_\mu(\phi, t)) \\ h_c(y_\mu(\phi, t)) \end{pmatrix}. \quad (3.23)$$

At a zero, exact row reduction gives

$$\det D_{(\phi,t)} \mathfrak{M}_\mu = s_\mu \chi_\mu, \quad s_\mu = dh_c(F_\mu), \quad \chi_\mu = \bar{L}_{c,\mu} D_\phi y_\mu. \quad (3.24)$$

The continuations of (1.9) satisfy $|s_\mu| \geq s_* > 0$ and $|\chi_\mu - \chi_0| < |\chi_0|/2$. Thus the phase/time operator is uniformly invertible and selects a unique C^2 source phase whose orbit crosses the original finite gate and stays on (3.20) to the algebraic end. Every truncated matching action satisfies exact moving-cut covariance.

Proof. The scaled outer graph extends to the comparison face $r = 0$. On a resolved K_1 block, uniform cone and bunching estimates give an overflowing center graph with fixed-extension uniqueness. Exact chart transitions transport it to both neighboring charts, and uniqueness makes the graphs agree on the overlaps. The final central trace is therefore an actual saturation of the same future-staying hypersurface, not a second asymptotic sheet. The resolved field and chart embeddings are C^2 in μ on the closed comparison family; graph-transform estimates yield (3.21) without a boundary trace loss.

The endpoint-normalized adjoint row is transported through this same atlas. At $r = 0$, (H_{Jost}) gives (1.10); continuity gives (3.22), so the directed endpoint conormal does not collapse. Separately, the normalized central target row converges to Dg_a , and the source flight converges in C^1 . Therefore (1.9) gives the two strict factors in (3.24). The parameter implicit-function theorem selects the source phase and flight time. Strict central first-hit

margins keep the selected orbit in the same gate component; invariance and local maximality then keep it on the future sheet.

Finally, all chart primitives are pullbacks of the single physical form λ_δ . For a first-hit map $P : C_0 \rightarrow C_1$ with flight time T_P , the ambient identity is

$$P^*\lambda_\delta - \lambda_\delta = d\mathcal{A}_P + T_P dH_{\text{ph}}. \quad (3.25)$$

On zero energy the final term vanishes on tangent vectors. Splitting the same finite orbit integral proves exact composition and moving-cut covariance. $\square$

3.3. Reference-tail finite parts. Fix a small outer cut $z = z_*$ on the matched sheet and use $Q = z^{-2}$. On zero energy, write the stable graph label as b and let $b_{\text{ref}}(Q; \mu)$ be the reference orbit selected by $b(z_*^{-2}) = 0$. Abbreviate, at $z = Q^{-1/2}$,

$$\Gamma = \Gamma_\mu(z, 0, b), \quad \text{Chi} = \text{Chi}(z, 0, b, \Gamma; \mu), \quad \Pi = \delta \text{Chi} + \Gamma + b, \quad W = \Gamma - b.$$

Then $dQ/d\tau = 2\Pi$ and the exact scalar reduced equation is

$$\frac{db}{dQ} = G(Q, b; \mu) = \frac{1}{2\Pi} \left[-b + \frac{z^2}{2} \{ -\delta^2 \epsilon (1 - az) + 2\delta \text{Chi} \Pi + \Pi + \Pi W \} \right]. \quad (3.26)$$

In particular,

$$\partial_b G(\infty, 0; \mu) = -\frac{1}{2\delta q_*(\epsilon)} < 0. \quad (3.27)$$

Hence any actual tail and the reference tail at the same Q differ exponentially:

$$\left| D_{b_0}^i D_\mu^j \{ b(Q; b_0, \mu) - b_{\text{ref}}(Q; \mu) \} \right| \leq C e^{-\eta(Q-Q_*)}, \quad i + j \leq 3, \quad j \leq 2. \quad (3.28)$$

The physical length and action densities in this coordinate are exactly

$$\begin{aligned} \mathcal{T}(Q, b; \mu) &= \frac{\delta}{2\Pi(Q, b; \mu)} Q^{-1/2}, \\ \mathcal{A}(Q, b; \mu) &= -\frac{\text{Chi}(Q, b; \mu)^2}{2\Pi(Q, b; \mu)} Q^{3/2} + \frac{\epsilon \Pi(Q, b; \mu)}{2} Q^{-1/2}, \end{aligned} \quad (3.29)$$

where Π and Chi are the positive outer momentum and energy root.

Proposition 3.3 (Algebraic finite parts). *Under Hypothesis 1.1, the complete reference integrals*

$$C_{\text{x}, \mu}(Q) = \int_{Q_*}^Q \mathcal{T}(q, b_{\text{ref}}; \mu) dq, \quad C_{\text{A}, \mu}(Q) = \int_{Q_*}^Q \mathcal{A}(q, b_{\text{ref}}; \mu) dq \quad (3.30)$$

have the leading behavior (1.14). The differences between an actual tail and the reference tail are integrable after every mixed derivative of total order at most two, and in fact after one spare state derivative. Consequently (1.13) exists, composes with the whole matched arrival patch, and satisfies exact cut, admissible coordinate, reference, and exact-gauge covariance.

Proof. The negative rate (3.27) gives a contraction on an exponentially weighted half-line. Differentiating its fixed-point equation three times gives (3.28). Since Π is uniformly positive, the mean-value formula applied to (3.29) gives integrable bounds of the form

$$\begin{aligned} |D^{\leq 3}(\mathcal{T} - \mathcal{T}_{\text{ref}})| &\leq CQ^{-1/2}e^{-\eta(Q-Q_*)}, \\ |D^{\leq 3}(\mathcal{A} - \mathcal{A}_{\text{ref}})| &\leq C(1+Q)^{3/2}e^{-\eta(Q-Q_*)}. \end{aligned} \quad (3.31)$$

Dominated convergence proves the finite parts and their derivatives. On the reference orbit, $\Pi = \delta q_* + O(Q^{-1/2})$ and $\text{Chi} = q_* + O(Q^{-1/2})$; inserting these in (3.29) and integrating proves (1.14). Because (3.30) uses the complete field-dependent tail, all lower powers and logarithms are subtracted automatically; no finite Taylor polynomial is being renamed a finite part.

For every finite Q , ordinary spatial-length and line-integral additivity holds before subtraction. The same single reference term appears on both sides, so the equality survives as $Q \rightarrow \infty$. Event-free section slides and exact changes $\lambda_\delta \mapsto \lambda_\delta + d\psi$ add endpoint coboundaries. Exponential flatness makes admissible coordinate and reference changes converge to finite endpoint corrections. This proves all stated covariances. $\square$

4. EXHAUSTIVE FIRST EVENTS AND STATIONARY SPATIAL PATTERNS

4.1. Minimal replacement of the two finite gates. We now return to the compact central event block. The construction retains the entire finite physical block of Proposition 2.1. Only two local refinements are made.

First, the existing transverse algebraic carrier is subdivided by the internal label that selects the matched sheet from Proposition 3.2. The carrier remains the physical event surface; the internal label supplies the terminal conormal but is not assigned a new hit time. Second, an event-free slide inside the protected pole outcome replaces its old gate by the actual $x = 10$ carrier in (2.7); that carrier is then partitioned into the pole aperture and its complement. Strict earlier-event and flow-buffer margins keep this replacement away from every competing event. Again no new event time is introduced.

The resulting grouped event arrangement contains the four source-cell boundaries; the original outgoing and homoclinic-return faces; the algebraic carrier, internal label, and patch faces; the $x = 10$ pole carrier and aperture faces; the nested return, target-sign, stable-cut, and return-box faces; and the original finite list of competing-time rows. All of these are pulled back through the same signed limiting source template before clean incidence and first-event order are tested. In the labels below, *out* is the named outgoing lateral boundary and *rbox* is the named return-box lateral boundary.

Proposition 4.1 (Compact-family first events). *Assume Hypothesis 1.1 and the constructions of Propositions 2.1, 2.2, 3.2, and 3.3. There is a finite marked cover $V_i \in U_i$ of (1.11), an intrinsic residence threshold T_* , an integer N , and local signed source collars $\Sigma_{N,i}$ with the following properties.*

In each marking, $\Sigma_{N,i}$ is the disjoint union

$$\begin{aligned} \Sigma_{N,i} = & \bigsqcup_{\substack{\sigma, \sigma' \in \{+, -\} \\ n \geq N}} \mathcal{V}_{\sigma, n}^{\sigma'} \sqcup \bigsqcup_{\substack{\sigma \in \{+, -\} \\ n \geq N}} \mathcal{I}_{\sigma, n} \\ & \sqcup \bigsqcup_{\substack{\mathbf{t} \in \{\mathbf{a}, \mathbf{p}, \mathbf{out}, \mathbf{rbox}\} \\ \sigma \in \{+, -\}, n \geq N, \ell}} E_{\mathbf{t}, \sigma, n, \ell}, \end{aligned} \quad (4.1)$$

where every union ranges only over nonempty connected components. Every point has one local winding, current sign, component label, and first event; every return has one next sign. The stable cut $\mathcal{I}_{\sigma, n}$ is an exit face and is not counted as either adjacent return. Both target-sign returns are nonempty for every $n \geq N$, and the anchored pole and algebraic components are nonempty for every $n \geq N$.

The displayed identity is a stratified disjoint union of connected relative interiors, not a union of open sets of one dimension. In particular, an algebraic terminal stratum is the trace of a cooriented hypersurface, whereas the pole aperture has a top-dimensional relative interior. Their common boundaries occur as separately named lower-dimensional strata.

For $a_* = (\sigma, n, \sigma')$, the completed return map has a cross form

$$x' = f_{a_*}(x, y'), \quad y = g_{a_*}(x, y') \quad (4.2)$$

on fixed rectangles. Its contraction is uniformly less than one, and for each sign pair there are limiting cross forms such that

$$\|f_{a_*} - f_{\mu, \sigma, \sigma', \infty}\|_{C_{z, \mu}^2} + \|\tilde{g}_{a_*} - \tilde{g}_{\mu, \sigma, \sigma', \infty}\|_{C_{z, \mu}^2} \leq C(1+n)^3 e^{-\kappa n}, \quad (4.3)$$

where

$$\tilde{g}_{a_*} = g_{a_*} / \varepsilon_{\mu, a_*}, \quad \varepsilon_{\mu, a_*} = \nu_{N, i}^{-1} \exp\left(-\frac{2\pi\alpha_\mu}{\beta_\mu} n\right). \quad (4.4)$$

On overlaps, the chartwise partitions have a finite common refinement and describe the same physical first-event relation on Σ_{T_*} ; their raw winding labels differ at most by bounded integer recoding.

Proof. Every new pole or algebraic face is an actual compact physical first hit with a strict speed, earlier-event exclusion, flow-domain buffer, and anchor-to-boundary margin. The carrier/label distinction above preserves the old finite list of competing hit times. Add the new conormals to the old clean event table and take one finite minimum of all normalized nonzero ranks and strict gaps. Compact first-hit stability preserves the incidence complex and physical order; near a genuine tie the nonzero tie conormal and the fixed priority replace any false time-gap assertion. The old finite block was already exhaustive, and the two operations only subdivide its algebraic and pole outcomes. Hence there is no residual component.

Apply the local conditional compact-family transfer proposition from the companion proof archive separately on each U_i . Its passage/matching inputs are precisely the frozen modules in (H_{ret}) ; its event input is the just-verified

physical arrangement, and its overlap input is the already constructed finite marked atlas. The proposition transfers these data to the compact family; it does not create or assume the application event census. Refine the cover so that there is a common rate

$$\varkappa < \varkappa_i^+ < \frac{2\pi \inf_{\mu \in U_i} \alpha_\mu}{\sup_{\mu \in U_i} \beta_\mu}. \quad (4.5)$$

This order of infimum and supremum is what supplies a uniform passage contraction on the cover member. The winding selector and clean event stability then give (4.1)–(4.3). The pole and algebraic anchors have fixed interior component labels, so their selected components persist for every large winding.

Finally, residence time and winding differ by a uniformly bounded amount after multiplication by $2\pi/\beta_\mu$. Choose the final N so that every displayed branch lies in the intrinsic collar Σ_{T_*} . On marking overlaps, exact section changes preserve the physical event and change the angular lift by a bounded integer. The finite marked-atlas descent therefore identifies the physical relation without asserting a global exact chart or a global integer alphabet. $\square$

4.2. Length, action, and exact composition. For a return branch, write

$$\iota_{\mu, a_*}(x, y') = (x, g_{a_*}(x, y'))$$

for the source inclusion. The physical clock factor in (2.4) and the saddle passage give

$$L_{a_*} = \frac{2\pi r}{\epsilon^{1/4} \beta_\mu} n + \hat{L}_{a_*}, \quad \mathcal{A}_{a_*} = \int_{a_*} \lambda_\delta, \quad (4.6)$$

and, for each sign pair,

$$\left\| \hat{L}_{a_*} - \hat{L}_{\mu, \sigma, \sigma', \infty} \right\|_{C_{z, \mu}^2} + \left\| \mathcal{A}_{a_*} - \mathcal{A}_{\mu, \sigma, \sigma', \infty} \right\|_{C_{z, \mu}^2} \leq C(1+n)e^{-\varkappa n}. \quad (4.7)$$

On a terminal branch, composition with the one-spore-derivative pole or algebraic potential gives the rate $C(1+n)^3 e^{-\varkappa n}$ after two derivatives.

For a finite first-hit branch P_γ , the ambient identity (3.25) restricts on zero energy to

$$P_\gamma^*(\lambda_\delta|_{\Sigma_\gamma^+}) - \lambda_\delta|_{\Sigma_\gamma^-} = d\mathcal{A}_\gamma. \quad (4.8)$$

If two branches compose, then

$$\mathcal{A}_{\gamma_2 \circ \gamma_1} = \mathcal{A}_{\gamma_1} + \mathcal{A}_{\gamma_2} \circ P_{\gamma_1}. \quad (4.9)$$

At either end, (4.9) is imposed at a finite cutoff; only the terminal segment receives its counterterm; the cutoff limit is taken last. This proves item 3 of Theorem 1.2 and prevents a terminal subtraction from being counted twice.

4.3. Coding and return to the PDE.

Completion of the proof of Theorem 1.2. The return rectangles in Proposition 4.1, their full sign adjacency, the completed cross forms, and the strict first-exit partition satisfy (H_{ret}) . The resulting two-sided edge shift on (1.15) is therefore conjugate to the trapped return set. The roof and action potentials have summable variations by (4.3) and (4.7). A contraction on the product of the rectangles for a primitive cyclic word gives one periodic spatial orbit. Summing (4.6) and using the exact cocycle gives (1.18) and the corresponding closed-action formula.

For a finite word $a_0 \cdots a_{m-1}$, use the completed section coordinates in which the selected primary unstable-image trace is $x = 0$ and the local stable trace is $y = 0$. Solve

$$x_{j+1} = f_{a_j}(x_j, y_{j+1}), \quad y_j = g_{a_j}(x_j, y_{j+1}), \quad x_0 = 0, \quad y_m = 0. \quad (4.10)$$

The product-box map is a contraction, hence has a unique solution. Internal edges lie in the regular part of the completed sections; the first boundary condition puts the backward orbit on the selected unstable manifold and the last puts the forward orbit on the local stable manifold. The resulting orbit is homoclinic to the homogeneous state and makes the prescribed finite sequence of high-winding visits. Words of arbitrary finite length therefore give multipulse profiles. The usual inverse-limit construction gives every bi-infinite coded orbit; nonperiodic itineraries remain nonperiodic because the first-event itinerary is injective on the trapped set.

All coded orbits remain in the compact saddle block and finite global flight tube, and branch times have a positive lower bound, so no finite spatial-time accumulation occurs. Applying the exact inverse transformation (2.1) gives a complete smooth physical orbit of (1.2). Eliminating p and q yields

$$0 = v - f(u) + du_{xx}, \quad 0 = \epsilon(a - u) + v_{xx}, \quad (4.11)$$

which is the stationary form of (1.1). Finite marked-atlas covariance identifies the physical profile, closed period, and closed action on overlaps, although its raw integer word may be recoded. This proves all remaining items of Theorem 1.2. $\square$

4.4. Interpretation boundary. The word “dynamics” in the title refers to spatial dynamics in the independent variable x . A stationary spatial shift, spatial entropy, or aperiodic stationary profile is not temporal chaos for the parabolic PDE. The theorem also does not establish temporal spectral or nonlinear stability, connection to a branch selected from the Turing bifurcation, canard tracking along attracting and repelling slow manifolds, or experimental observation. Those are separate questions about objects whose stationary existence is established here.

APPENDIX A. PROOF ARCHIVE AND STATEMENT CROSSWALK

The main text organizes the argument by mathematical obstruction rather than by the historical labels V1–V7. The complete coordinate calculations, polyhomogeneous recursions, cone estimates, matching-operator bounds, mixed jet bookkeeping, and event-face tables are included in the same immutable release as the following versioned proof files:

label	mathematical output	proof location
V1	physical exact Hamiltonian and central scaling	HAMILTONIAN_CHECK, MODEL_AND_CENTRAL_CHART
V2	central homoclinic, passage, and finite events	CENTRAL_CONTINUATION
V3	positive pole and pole finite part	POSITIVE_POLE_FINITE_PART
V4	outer future-staying graph	OUTER_FUTURE_STAYING
V5	$K_2 \rightarrow K_1 \rightarrow$ outer matching	CENTRAL_OUTER_MATCHING
V5A	algebraic length/action finite parts	OUTER_ALGEBRAIC_FINITE_PART
V6–V7	physical first events and stationary coding	TWO_END_RETURN_EXIT_AND_PDE

The filenames are shortened in the table; their full paths and exact source versions are listed in the provenance supplement. No proof-bearing step of the main theorem is delegated to the numerical Issue #7 candidate box.

APPENDIX B. DATA AND EVIDENCE AVAILABILITY

The public release

<https://github.com/h-lu/rfsn-ii-positive-parameter-pde/releases/tag/vdp-companion-v1>

contains the paper sources, companion proof archive, frozen RFSN-II source manuscript, interval certificates, validation source, environment lock, and replay manifest. The separate supplement records SHA-256 identities, evidence roles, and the fail-closed replay command; exact frozen theorem labels and locations are in the versioned import notes cited there. The release makes Hypothesis 1.1 externally inspectable; it does not turn an imported certificate into an independent second-machine replay.

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